

Revisiting Learning Paradigms for Multimedia Data Generation

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ABSTRACT

With the development of deep learning, multimedia data generation (e.g., image generation, audio synthesis, and video generation) has attracted a lot of attention. Deep learning methods for data generation usually build a mapping from source condition X to target data Y . The target Y (e.g., image, speech, music, video) is usually high-dimensional and complex, and contains rich information not exist in source data, which hinders the effective and efficient learning on the source-target mapping. Representation learning has achieved rapid progress in the past decade, which is beneficial for data understanding tasks. However, traditional representation learning cannot address the challenges faced by multimedia data generation tasks. This tutorial revisits the learning paradigms for data generation and introduces a paradigm called *regeneration learning* that can improve the effectiveness and efficiency of multimedia data generation. We show that a variety of tasks in multimedia data generation (e.g., image generation, speech synthesis, music composition, video generation) can benefit from this regeneration learning paradigm, and a lot of recent popular data generation models (e.g., DALL-E 1/2, Stable Diffusion, AudioLM, NaturalSpeech 2, MusicLM) can be covered by this learning paradigm.

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1 CHALLENGES OF DATA GENERATION

Multimedia data generation suffers from one-to-many mapping and spurious correlation issues:

- First, for the generation tasks where the target Y contains much more complex information than the condition X , the generation models face severe one-to-many mapping problem (ill-posed or ill-condition problem) [2], which increases the learning difficulty. For example, in image generation, a class label “dog” can correspond to different images that contain dogs. Incorrect modeling on the ill-posed problem could result in overfitting on the training set and poor generalization on the test set.
- Second, for the generation tasks (e.g., speech synthesis [15], talking face synthesis [14]) where Y contains no significantly more or less information than X , there are two situations: 1) the mapping

between X and Y is not one-to-one, e.g., multiple speeches with different speaking rates can correspond to one text sequence in text-to-speech synthesis; 2) there are some spurious correlations between X and Y , e.g., the speaking timber in source speech has no correlation with head pose in target video for talking face synthesis, and most information in target melody has no correlation with source lyric in lyric-to-melody generation [6]. Fitting these spurious correlations can be harmful for generation in inference.

2 WHAT IS REGENERATION LEARNING AND WHY

Ideally, as long as generative models (e.g., GANs [4], VAEs [7], autoregressive models, normalizing flows [10], diffusion models [5]) are powerful enough, they can fit any complex data distribution. However, in practice, they cannot model the complex distribution and one-to-many mapping well due to many reasons, such as too complicated data mapping, too heavy computation cost, and data sparsity, etc. In analogy to data understanding tasks, although learning a powerful model (e.g., CNN or Transformer) to directly classify source data into target labels would ideally achieve very good accuracy, it still suffers from low accuracy, and some advanced representation learning methods such as large-scale self-supervised pre-training can greatly boost the accuracy.

In this tutorial, we revisit and present a learning paradigm called *regeneration learning* [13] (joint work between Microsoft researchers and Yoshua Bengio¹) for data generation, which leverages an intermediate representation of target Y to bridge the information mismatch between X and Y . There are three steps in regeneration learning: 1) Convert target Y into an abstractive/representative version Y' , as shown in Table 1; 2) Learn a model to generate Y' from source data X ; 3) Learn another model to generate Y from Y' . Since there are two steps for generation (first generate Y' and then generate Y , i.e., regenerate), this learning paradigm is called regeneration learning. Regeneration learning has several advantages: 1) $Y' \rightarrow Y$ mapping can be learned in a self-supervised way (Y' is obtained from Y) and can be empowered by large-scale pre-training that is similar to that in traditional representation learning for data understanding tasks; 2) $X \rightarrow Y'$ mapping will be less one-to-many than $X \rightarrow Y$ since Y' is a compact/representative version of Y .

3 REGENERATION LEARNING VS REPRESENTATION LEARNING

Regeneration learning extends the concept of representation learning to data generation tasks, and can be regarded as a *counterpart* of traditional representation learning, since 1) regeneration learning handles the abstraction (Y') of the target data Y for data generation

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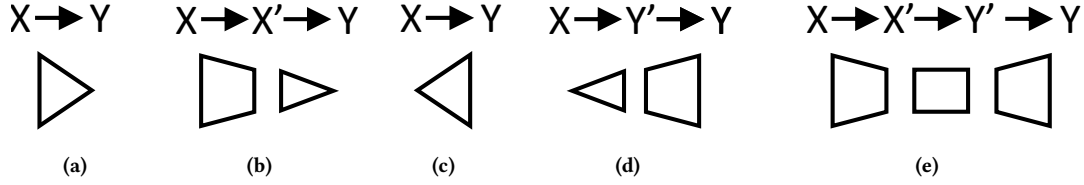
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¹Xu Tan, Tao Qin, Jiang Bian, Tie-Yan Liu, Yoshua Bengio. Regeneration Learning: A Learning Paradigm for Data Generation. arXiv 2023.

Formulation	Category	Method	Data Conversion ($Y \rightarrow Y'$)
Basic	Explicit	Fourier Transformation	Speech/Image (e.g. Wave $\rightarrow$ Spectrogram)
		Music Analysis	Music (MIDI $\rightarrow$ Chord/Rhythm)
		3D Image Analysis	Image (Face to 3D Co-efficient)
		Down Sampling	Speech/Image (e.g., $256 \times 256 \rightarrow 64 \times 64$)
	Implicit	VAE/VQ-VAE/VQ-GAN	Image/Speech/Text ($Y \rightarrow Z$)
Extended	Factorization	AR	Image/Speech/Text ($Y \rightarrow Y_{1:t}$)
	Diffusion	DDPM	Image/Speech/Text ($Y_0 \rightarrow Y_t$)
	Latent Diffusion	VAE + DDPM	Image/Speech/Text ($Y \rightarrow Z_0, Z_0 \rightarrow Z_t$)

Table 1: Different methods for $Y \rightarrow Y'$ conversion.Figure (a): Presentation ($X \rightarrow Y$). Figure (b): Representation ($X \rightarrow X' \rightarrow Y$). Figure (c): Generation ($Y \rightarrow X$). Figure (d): Regeneration ($X \rightarrow Y' \rightarrow Y$). Figure (e): Representation + Regeneration ($X \rightarrow X' \rightarrow Y' \rightarrow Y$).

Paradigm	Original	Compact	Self-Supervised Learning	Easy Mapping
(b) Representation Learning	X	X'	$X \rightarrow X'$	$X' \rightarrow Y$
(d) Regeneration Learning	Y	Y'	$Y' \rightarrow Y$	$X \rightarrow Y'$
(e) Combination	X, Y	X', Y'	$X \rightarrow X', Y' \rightarrow Y$	$X' \rightarrow Y'$

Figure 1: Comparison between regeneration learning and traditional representation learning on $X \rightarrow Y$. Figure (a): Data understanding tasks, where X contains more information than Y . Figure (b): Data understanding tasks with representation learning, where X' is a more compact version of X . Figure (c): Data generation tasks, where Y contains more information than X . Figure (d): Data generation tasks with regeneration learning, where Y' is a more compact version of Y . Figure (e): Data understanding/generation tasks with the combination of representation and regeneration learning, where X' and Y' contain a comparable amount of information and are compact versions of X and Y respectively.

Model	X	Y	Y'	$X \rightarrow Y'$	$Y' \rightarrow Y$
DALL-E	Text	Image	Visual Token	AR	VQ-VAE Decoder
DALL-E 2	Text	Image	CLIP Latent	AR/Diffusion	Diffusion
Stable Diffusion	Text	Image	VAE Latent	Diffusion	VAE Decoder
AudioLM	Audio Prompt	Audio	Semantic/Acoustic Token	AR	VQ-VAE Decoder
MusicLM	Text	Music	MuLan/Semantic/Acoustic Token	AR	VQ-VAE Decoder
NaturalSpeech 2	Text	Speech	VQ-VAE Latent	Diffusion	VQ-VAE Decoder
GPT-3/ChatGPT	Text Prompt	Text	Text Prompt	Chain-of-Thought Prompting	

Table 2: Some recent data generation models that leverage regeneration learning.

while traditional representation learning handles the abstraction (X') of source data X for data understanding; 2) both the processes of $Y' \rightarrow Y$ in regeneration learning and $X \rightarrow X'$ in representation learning can be learned in a self-supervised way (e.g., pre-training); 3) both the mappings from X to Y' in regeneration learning and from X' to Y in representation learning are simpler than the direct mapping from X to Y . Figure 1 shows the comparison between representation learning and regeneration learning.

4 APPLICATIONS

Regeneration learning can be applied to a variety of multimedia data generation tasks (e.g., speech synthesis, music composition, image

generation, and video generation) and can cover the pipeline used in recent popular data generation models, such as DALL-E 1/2 [8, 9], Stable Diffusion [11], AudioLM [3], MusicLM [1], NaturalSpeech 2 [12]. We show how recent popular data generation models are covered by regeneration learning in Table 2.

5 RESEARCH OPPORTUNITIES

We list some research opportunities to make regeneration learning more powerful to solve a variety of data generation tasks, mainly from three perspectives: 1) how to get Y' ; 2) how to learn the mapping $Y' \rightarrow Y$ in a self-supervised way; 3) how to reduce the training-inference mismatch in regeneration learning pipeline.

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